

Constraints on kinship semantics reflect cognition above cultural practice

Maisy Hallam (maisy_hallam@eva.mpg.de)^a, Simon Kirby^b, Fiona M. Jordan^c, Kenny Smith^b

^aDepartment of Linguistic and Cultural Evolution Max Planck Institute for Evolutionary Anthropology

^bCentre for Language Evolution University of Edinburgh

^cDepartment of Anthropology and Archaeology University of Bristol

Abstract

Languages encode similar semantic features in their kinship terminologies, the set of words denoting family members, but different languages seem to prioritise different distinctions. Is any configuration of kinship semantics possible, or are there commonalities across languages in which distinctions are encoded? We use computational methods to identify clear cross-linguistic tendencies in how six semantic features are encoded in kinship terms. Across 1242 languages, all encode information on Generation, while other features like Gender or Seniority are less consistently encoded. Importantly, the features are encoded in a predictable order of importance. We find that this tendency mirrors participants' behaviour in a linguistic categorisation experiment, suggesting kinship semantics reflect cognitive biases active at the level of the individual. Additionally, despite claims that kin terms are a straightforward reflection of social practice, we find varying evidence for this; kin term semantics systematically correspond with patterns of descent, but not marriage norms.

1. Introduction

Kinship terms, the set of words that denote family relationships, are strikingly diverse across the world's languages (Murdock, 1949; Greenberg, 1990; Nerlove & Romney, 1967; Kemp & Regier, 2012). Different languages incorporate different aspects of meaning into their kinship terminology: for example, an English speaker will distinguish their *brother* from their *sister* by their gender, while an Indonesian speaker distinguishes their *kakak* 'older sibling' from their *adik* 'younger sibling' by their seniority (i.e. by their relative age). Meanwhile, a Hawaiian speaker does away with the lineality distinction between siblings and cousins entirely, using *kaikuahine* to group female siblings with female cousins and *kaikua'ana* to group male siblings with male cousins.

In this paper, we explore the extent of this variation and the cognitive and communicative factors that shape it. Building on an established finding that kinship terms are designed for efficient communication (Kemp & Regier, 2012), we apply computational methods to a database of kinship terminologies from 1242 languages and discover a shared hierarchy of which features are encoded and how strongly. We also address the cognitive and cultural mechanisms that underpin this typological observation. Using a simple linguistic categorisation experiment, we show that participants prefer to extend kin terms to individuals who share

semantic features that rank higher on this cross-linguistic hierarchy, suggesting it is rooted in cognitive biases active at the level of the individual. Additionally, we show that the ranking of certain semantic features sometimes reflects the social practice of the linguistic community, for some kinds of social practice.

1.1. A topographic relationship between form and meaning

The potential variation in kin term semantics is vast, but it has been argued that – as in other linguistic domains – there are constraints on this variation. Lexical semantics have been repeatedly shown to support efficient communication (Kemp et al., 2018; Gibson et al., 2019) in domains such as colour (Zaslavsky et al., 2018), numerals (Y. Xu, Liu, & Regier, 2020), and indeed, kinship (Kemp & Regier, 2012). Natural language lexicons in these domains optimally trade-off competing pressures to be simple, yet informative; they seem to be designed so as to permit the formation of compressed mental representations, while at the same time facilitating precise communication of important conceptual distinctions.

One reflex of this representational efficiency is that similar wordforms tend to have similar meanings: the mapping between form and meaning is **topographic** (Brighton & Kirby, 2006). Topographicity manifests in several ways, both in the general lexicon and in kinship terminology. When two concepts are expressed with identical word forms (i.e., when they are colexified), they are often highly similar in meaning (Y. Xu, Duong, et al., 2020), and experimental work shows that people prefer to colexify similar meanings unless this is detrimental to communication (e.g., if two similar meanings need to be distinguished frequently: Karjus et al., 2021). For example, the term *uncle* in English expresses two highly similar concepts, ‘mother’s brother’ and ‘father’s brother’, and extensions of *uncle* to non-kin (e.g. to parents’ friends) preserve the notion of a kin-like, older, male individual. Even where word forms are not *identical*, there is also a subtle tendency for words with similar meanings to have similar forms (Monaghan et al., 2014; Dautriche et al., 2017). One example of this is phonaesthemes, a recurring mapping between sound segments and meanings beneath the lexical or morphemic level (Firth, 1930). For example, the *gl-* onset in English tends to appear in words that have to do with light, such as in *glisten*, *glitter*, *glimmer* and *glow* (Firth, 1930; Bloomfield, 1933). Finally, there may be a correspondence between form and meaning due to compositionality, a pervasive feature of language where the meaning of complex expressions are composed of the meanings of their parts. For example, in Swedish, the terms *mor* ‘mother’ and *far* ‘father’ recombine to make a four-way distinction among grandparents: *mormor* ‘mother’s mother’, *farmor* ‘father’s mother’, *morfar* ‘mother’s father’ and *farfar* ‘father’s father’.

Previous research has treated the topographic mapping between forms and meanings as a research finding in itself, noting that meaning-form mappings tend to be structured according to this principle across the lexicon (Brighton & Kirby, 2006). We instead use this principle as a tool, a means to discover precisely which semantic content is encoded by shared form. We assume that similar forms tend to correspond to similar meanings, and ask what representation of meaning-similarity maximises this correspondence. This allows us to (1) reconstruct how strongly different kinds of semantic content are encoded in the kinship terminology

of *particular* languages and (2) identify common trends in which semantic content is encoded in kin terms *across* languages.

1.2. Topographicity reveals the semantic structure of kinship

Across the world’s languages, distinctions between kin terms are made with primarily six semantic features (Kroeber, 1909): **Generation** (the generation an individual belongs to), **Gender** (their gender), **Lineality** (whether they are in your direct line of descent), **Gender of Connecting Relative** (GCR; whether they are connected to you by a male or female relative), **Seniority** (whether they are older or younger than some counterpart), and **Speaker Gender** (the gender of the person using the term). All features are distinctive in some languages, but not all features are distinctive in all languages; for example, English distinguishes kin by Generation (*mother* vs. *grandmother*), Gender (*brother* vs. *sister*) and Lineality (*father* vs. *uncle*), but not by GCR, Seniority or Speaker Gender. However, these latter features are encoded in other languages: we see GCR distinctions in Hindi (*mausi* ‘mother’s sister’ vs. *phūphī* ‘father’s sister’), Seniority distinctions in Indonesian (*kakak* ‘older sibling’ vs. *adik* ‘younger sibling’), and Speaker Gender distinctions in Tonga (*tokoua* ‘same sex sibling’ vs. *tuonga’ane* ‘opposite sex sibling’).

However, beyond identifying these core semantic features, anthropologists have hesitated to propose universal tendencies in the encoding of kinship semantics. In previous research, discussions of about kin term semantics have focused only on small subsets of the kinship tree, such as which distinctions are made among siblings (Nerlove & Romney, 1967; Jordan, 2011) or grandparents (Sheard et al., 2020). In general, it is assumed that a system of kinship terminology is free to elaborate any or all semantic features to any extent (cf. Jones, 2010). This is similar to what we see in other semantic domains such as sensory perception, where languages are mostly free to vary how richly they encode sight, smell, touch and taste (with very few constraints, such as smell often being poorly encoded: Majid et al., 2018). Here, our novel contribution is to identify which of the six features is encoded and how strongly in the kinship terminologies of *individual* languages, and to identify the extent to which each feature is encoded *across* the world’s languages, by exploiting the presumed topographic relationship between shared form and shared semantic content.

To do this, we take a set of kin terms from a given language and measure how similar kin terms are (using inverse normalised edit distance; i.e. how much of the two strings are shared) and how similar their meanings are (a weighted sum of the semantic features they share). Our novel approach is to assume topographicity, i.e. we expect that there should be a strong correlation between wordform similarity and meaning similarity. Our model therefore seeks to optimise the correlation between these two measures by manipulating the measure of meaning similarity: if a semantic feature is frequently shared by individuals who share the same (or similar) kin terms (and not shared by individuals who have different kin terms) the model will upweight that semantic feature so that it gives a greater contribution to the measure of semantic similarity. Likewise, features which do not correlate with wordform similarity will be downweighted in the measure of semantic similarity. This process leaves us with a set of optimal feature weights for each language — optimal in the sense that they are the most informative about which features determine which kin terms

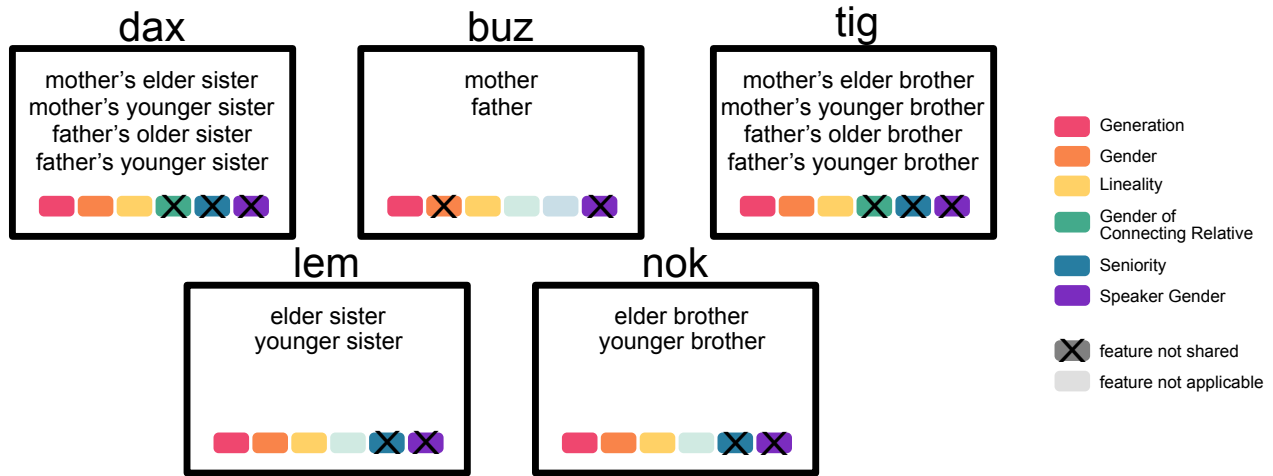


Figure 1: Schematic depicting how shared forms map to shared meanings in a miniature language. Each box represents a kin term, its contents the meaning of the kin term. Each kin term refers to meanings which share some features, and deviate on others: for instance, *dax* refers to kin who are all the same Generation (parents' generation), the same Gender (women), and the same Lineality (collateral), but they need not share GCR, Seniority, or Speaker Gender – this is the meaning of the term *aunt* in English. For this language, our model would give high weights to Generation, Gender, and Lineality, and a weight of 0 to GCR, Seniority, and Speaker Gender, as this would give the optimal correlation between shared forms and shared meanings. A set of weights that give, for example, a large proportion of weight to Seniority would produce a worse correlation, as several individuals who share a term do not share their Seniority value (e.g. *lem* and *nok*).

share all or part of their form. We interpret the weights as representing the extent to which each feature is encoded in the language's kinship terminology. A higher weight indicates that sharing this feature is highly correlated with sharing form, and a lower weight indicates that sharing that feature has little or no impact on sharing form.

We applied this model to kinship terminology systems from 1242 languages taken from Kinbank, a database of kinship terminology (Passmore et al., 2023). As expected, we found that individual languages differ in which semantic features they encode. However, we also found that the features differ in the extent to which they are encoded across languages, revealing a universal hierarchy for kin term semantics. At the top of the hierarchy we find semantic features that are encoded robustly and frequently across the world's languages (e.g. Generation), and at the bottom we find those that are encoded more weakly and rarely (e.g. Speaker Gender).

1.3. Cognitive and cultural mechanisms underpin the cross-linguistic distribution of kin semantics

What mechanisms might drive this pattern of semantic variation across languages? One possibility is that constraints on kinship semantics are shaped by biases in *cognition*: the individual-level preferences that people have when learning or using language are reflected in the typological distribution (Christiansen & Chater, 2008; S. Kirby et al., 2007, 2008, 2015; Smith, 2022; Culbertson et al., 2012; Culbertson & Newport, 2015). If so, we expect that people may be biased towards classifying kin by semantic features which are ranked high in the hierarchy rather than lower-ranked ones, and this cognitive salience makes them more

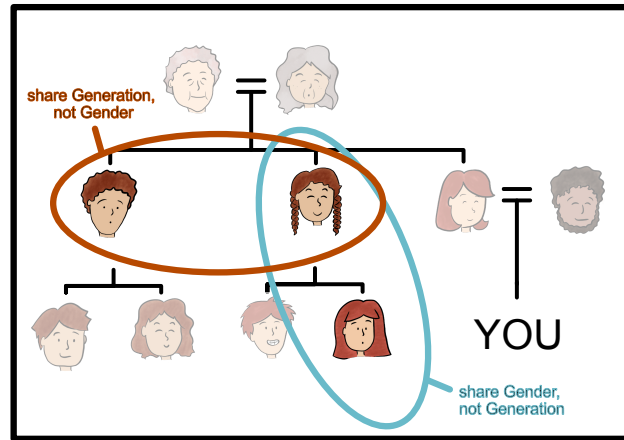


Figure 2: In the experiment, participants had to decide how to classify kin. They were given a kin term for one individual (e.g., the mother’s sister) and could choose to extend that term to one of two other highlighted kin. In this example, extending the term to the mother’s brother would indicate a preference for categorising by Generation, while extending to the mother’s sister’s daughter would indicate a preference for categorising by Gender. The three kin were identical on all other feature dimensions.

central and more widely used across languages. To investigate this, we ran a categorisation experiment where participants were tested on their preferences when classifying kin. Participants were shown a kin term for a particular relative in a novel language, and could choose to generalise that kin term to one of two other relatives (see Figure 2). These three relatives varied along a pair of feature dimensions, such that participants’ choice about how to extend the term would inform us which feature they were more likely to use as a cue to word meaning. For instance, when extending a kin term, do participants assume it applies to individuals of the same Generation, or the same Gender (all else being equal)? We find that people prefer to classify together kin who share higher ranked features on the hierarchy, suggesting that they are biased to see some features as better cues to shared category membership than others.

Variation in kinship semantics might also be driven by *cultural practice*, as has been routinely proposed in the anthropological literature for decades (e.g. Murdock, 1949; Coult, 1965; Goody, 1970; Rácz et al., 2020). We therefore also test the extent to which the kin term semantics of individual languages reflect the social practice of their communities, focusing on two predictions from the anthropological literature (Murdock, 1949; Goody, 1970): that kin terms reflect patterns of descent, and that kin terms reflect kin marriage practices. We find that the relative ranking of features can be predicted by a society’s patterns of descent – i.e. whether they trace ancestry and inheritance through the mother’s line, father’s line, or both lines – but *not* by their acceptance of cousin marriage. This suggests that some (but not all) social practices may influence communicative need strongly enough to shape a language’s kinship terminology – and that, within a particular language, the semantic hierarchy is malleable to such cross-culturally variable communicative pressures.

Taking these results together, we propose that while cross-linguistic commonalities in kin term semantics emerge from biases in human cognition that reflect the salience of particular features, variation between

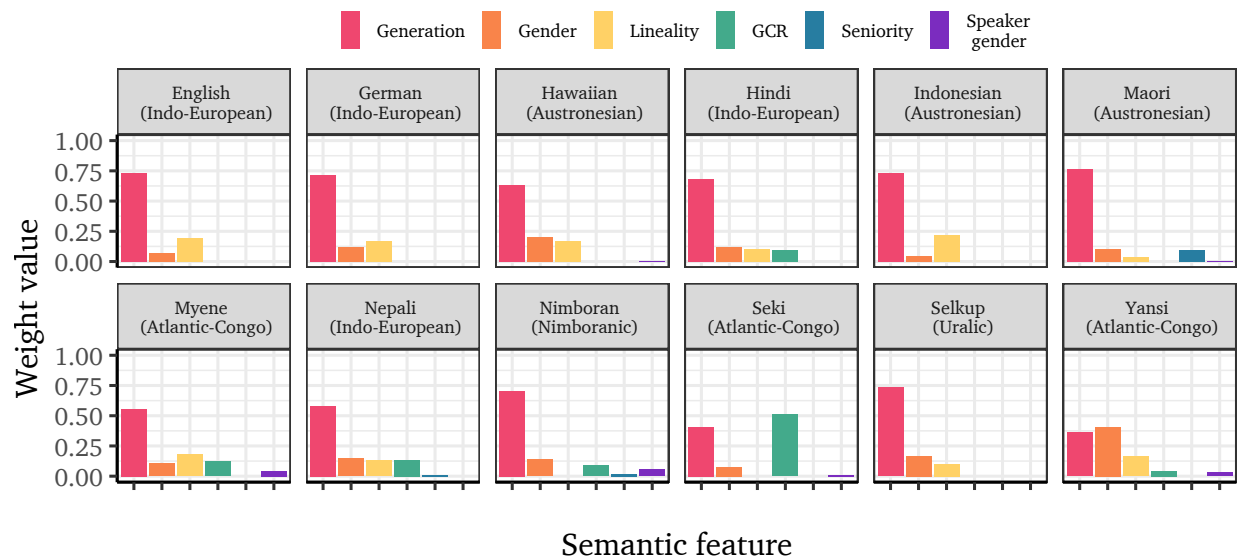


Figure 3: Final weights on each semantic feature for a random sample of languages (plus English). Each feature has a value between 0 and 1, which expresses how much that feature contributes to kinship semantics in that language. Languages differ from each other in terms of which features they encode and what weighting they receive. However, there are common tendencies: e.g. every language encodes Generation and Gender; Generation is usually the highest weighted, and Seniority and Speaker Gender receive very low weights (where they are encoded at all).

languages may also be a partial reflection of how useful those features are within a particular cultural niche.

2. Results

2.1. Kin term semantics are structured hierarchically

Under the assumption that similar wordforms will have similar meanings, our model outputs a weight value for each feature in each language. This value quantifies its relative importance for distinguishing between kin in that language. Figure 3 shows the output of our computational analysis of semantic feature weights for a sample of languages in our dataset, confirming that languages differ in which semantic features they encode and the extent to which they encode them. For example, in English, Generation has the highest weight, reflecting that individuals often share a term if they are in the same generation. Lineality is second, capturing how *grand-*, *son*, *daughter*, *mother* and *father* are re-used for individuals in Ego’s direct lineage. Last is Gender; individuals who share a term in English often share a gender (e.g. ‘mother’s sister’ and ‘father’s sister’ are both *aunt*). However, the weight is relatively small since sharing a gender is not a reliable predictor of form similarity in English (e.g. *mother* and *aunt* both denote women but share no part of their form). The remaining features receive a weight of 0, as they do not contribute to kin term distinctions in English.

Figure 4 shows the distribution of feature weights across all 1242 languages in our sample, revealing clear universal trends in the distribution of weights. For instance, Generation is the only feature that is invariably encoded in kinship terminology, while all other features are often weighted 0 (i.e., not encoded). In particular,

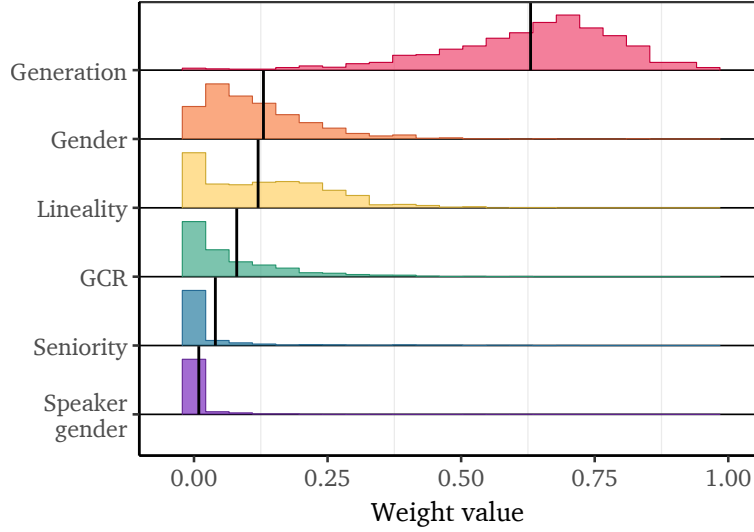


Figure 4: The hierarchy of semantic encoding across all languages in our dataset, represented as distributions of weights assigned to each semantic feature. Black lines indicate mean weight for each feature, and features are arranged in descending order by their mean value. Generation is generally weighted much higher than all other features, and rarely receives a weight of 0. Other than Generation, features are variably encoded, with Seniority and Speaker Gender the most often weighted 0, but there is nonetheless a clear hierarchy in how kin term semantics are encoded across the world’s languages: each successive feature has significantly lower weight than its higher-ranked neighbour.

Predictor	<i>b</i> (log-odds)	<i>SE</i>	<i>z</i>	<i>p</i>	95% CIs (log-odds)
Gender – Generation	-2.12535	0.03942	-53.92	<0.001	(-2.17643, -2.07427)
Lineality – Gender	-0.47488	0.04154	-11.43	<0.001	(-0.55630, -0.39326)
GCR – Lineality	-0.44978	-0.04277	-10.52	<0.001	(-0.53361, -0.36596)
Seniority – GCR	-0.68180	0.04322	-15.78	<0.001	(-0.76651, -0.59709)
Speaker Gender – Seniority	-0.19795	0.04312	-4.59	<0.001	(-0.28247, -0.11343)

Table 1: Beta regression model estimates predicting Weight by Semantic Feature using successive differences coding.

Seniority and Speaker Gender are rarely encoded at all. This suggests that not only do languages vary in what features they encode, but the features themselves vary in the extent to which they are encoded. To support this observation, we fit a mixed effects beta regression predicting the effect of Semantic Feature (sum-coded) on Weight with random intercepts for Language Family. Model comparison to a null model without the fixed effect for Feature indicated that Feature is an important predictor of Weight ($\chi^2 = 6287.3, df = 5, p < 0.001$); i.e. the weights on each feature differ systematically.

Additionally, Figure 4 suggests a typological hierarchy of how semantic features are encoded in kin terms: Generation is encoded the most strongly and often, and each successive feature is encoded less strongly and less frequently. To test for the presence of a reliable ordering of feature weights across languages, we fit a second beta regression model predicting Weight by successive Semantic Feature with random intercepts for Language Family. The full model output and contrasts are given in Table 1.

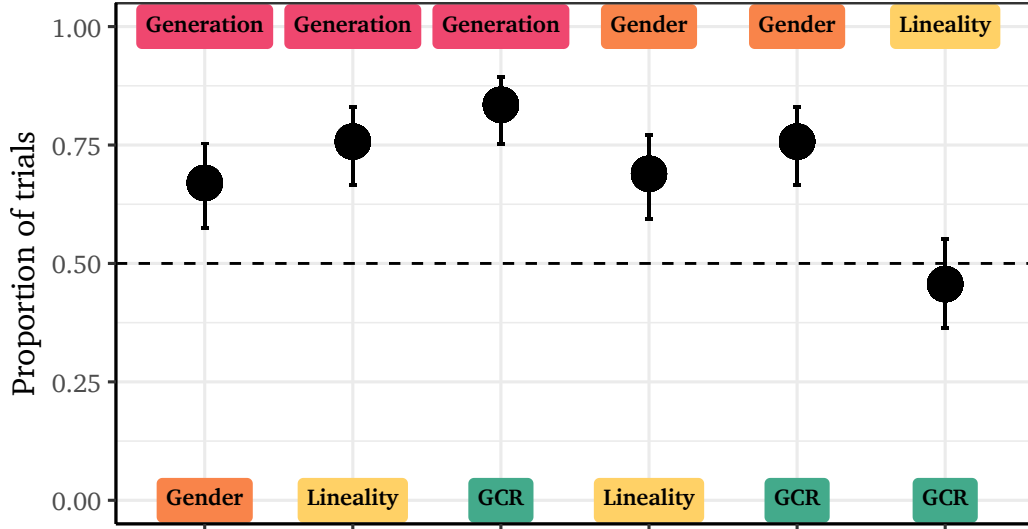


Figure 5: Proportion of trials in which participants selected the higher-ranked feature over its competitor. Error bars indicate 95% confidence intervals. Dotted line indicates chance at 50%. Participants are very likely to choose the higher ranked feature in all trials, and even more likely if the features are further apart on the hierarchy – suggesting their classification behaviour is aligned with the cross-linguistic hierarchy. However, they perform at chance on the Lineality-GCR trial, which may suggest that they do not have a preference between these two features, or perhaps that they struggle to identify what the relevant feature distinctions are in this trial type.

The model revealed a significant decrease in weight between each pair of successive contrasts, meaning that Generation receives a reliably higher weight than Gender, Gender than Lineality, and so on (all p -values < 0.001). However, effect sizes between each successive semantic feature in the hierarchy vary considerably, suggesting that the position of some features in the hierarchy is more robust than others: Generation is reliably the top ranked, but the relative order of Gender, Lineality, and GCR is more likely to vary between languages; Seniority and Speaker Gender are reliably weighted lower than all other features, but may vary in order between themselves. Based on this analysis, we propose the following cross-linguistic hierarchy: $\{\text{Generation}\} > \{\text{Gender, Lineality, GCR}\} > \{\text{Seniority, Speaker Gender}\}$, where brackets indicate multiple features which are similarly likely to be encoded in kinship terminologies.

2.2. Individual biases in classification reflect the cross-linguistic hierarchy

Having identified a cross-linguistic regularity in kin term semantics using natural language data, we tested experimentally whether this hierarchy is reflected in individual cognition. Do people prefer the higher-ranked features when determining whether a pair of kin should be referred to by the same term? Our participants completed a task where they had to choose to classify pairs of kin who differ minimally in the semantic features they share (see Figure 8). They were shown a kin term for one relative in an artificial language, and could choose to extend this term to one of two further relatives – one who shared a higher-ranked feature, or one who shared a lower-ranked feature. Participant behaviour in the experimental task was consistent with

the semantic hierarchy we identified in the cross-linguistic data (see Figure 4). An intercept-only logistic regression with by-participant random intercepts indicated that, across all trials, participants were more likely to classify kin who shared the higher-ranked of the two features ($\beta = 0.842, SE = 0.098, p < 0.001$). To illustrate this: in the trial example in Figure 8, the participant sees that the label *kawuka* is used to refer to the mother’s sister, and must choose whether to extend this label to the mother’s brother or mother’s sister’s daughter. Our participants preferentially extended the label *kawuka* to the mother’s brother, by the inference that he shares a higher-ranked feature (Generation) with the mother’s sister.

We also found that this effect gets stronger the greater the distance between the position of two features on the hierarchy. A model including fixed effects for the stepwise distance between features on the hierarchy shows that the higher ranked feature was chosen more often the greater the distance from its competitor (one step [model intercept]: $\beta = 0.44, SE = 0.126, p < 0.001$; two steps $\beta = 0.736, SE = 0.204, p < 0.001$; three steps $\beta = 1.232, SE = 0.295, p < 0.001$). For example, participants chose to classify by Generation over GCR (three steps) more reliably than they chose to classify by Generation over Gender (one step). However, Figure 5 suggests that while participants generally choose the higher ranked feature in each trial, they do not exhibit a preference in the Lineality-GCR trial. An intercept-only linear regression on just this trial type confirms that participants do not reliably choose the higher ranked feature in these trials ($\beta = -0.1752, SE = 0.1979, p = 0.376$). We consider this further in the Discussion.

Overall, our results indicate that there is a close correspondence between our semantic feature hierarchy, derived from the topographicity of kinship terminologies in the world’s languages, and the behaviour of participants extending kinship terms in a controlled experiment. Participants preferred to extend terms to kin who share features that are more common and more highly weighted cross-linguistically, meaning that kin sharing higher-ranked features were more likely to be assigned to a shared kin category.

2.3. *Inter-language variation reflects social practice*

Although our experiment results suggest that the hierarchy of kin term semantics is reflected in individual cognition, our cross-linguistic analysis also revealed a great deal of inter-language variation in the relative ranking of features: not all languages follow the semantic hierarchy strictly. It is commonly proposed that kinship terminology systems are shaped by social practice, particularly patterns of inheritance, marriage, or descent (e.g. Murdock, 1949). Cross-linguistic variation in the encoding of kin term semantics may therefore be driven by what information is relevant in particular cultural setting.

We tested two key claims from the anthropological literature about how social practice interacts with the relative ranking of each feature. The first claim is that unilineal societies, which trace ancestry and inheritance through either the maternal or paternal line, privilege distinctions between maternal and paternal kin. Bilateral societies, which trace both parents’ lineage, should accordingly distinguish kin in the direct line of descent from collateral kin (Murdock, 1949; Coult, 1965). If this were the case, we would expect unilineal societies to rank GCR higher, reflecting the importance of distinguishing kin in the matri- or patriline, whereas bilateral societies should rank Lineality higher, reflecting the importance of the direct line.

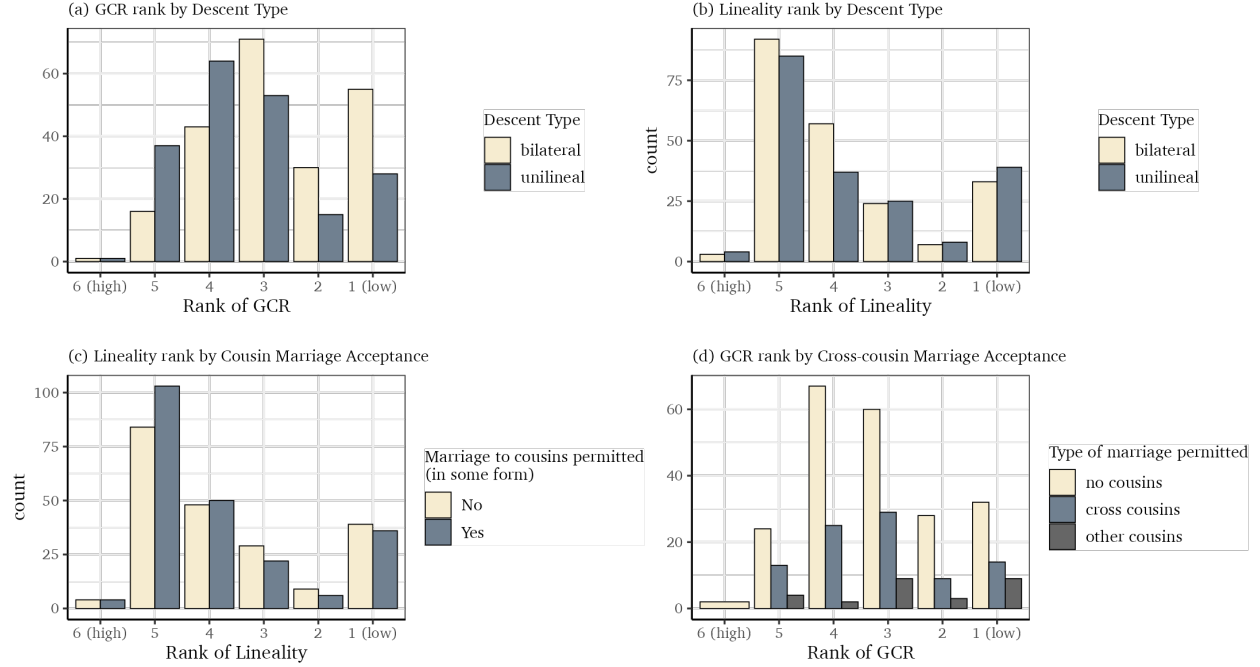


Figure 6: Distribution of semantic feature rankings for societies by descent type (top), and cousin marriage practices (bottom). A ranking of 6 means the feature is weighted highest of all features, while a ranking of 1 means it is weighted the lowest. Bilateral societies tend to rank Lineality higher than unilineal societies and GCR lower than unilineal societies, in line with our predictions. Societies who permit any cousin marriage tend to rank Lineality slightly higher, but this effect is not significant. Societies who permit cross-cousin marriage do not rank GCR higher than societies that permit no cousin marriage. In all, we suggest that descent patterns do predict kin term semantics, but not cousin marriage practices.

Both predictions are supported by our data. An ordinal regression model predicting the effect of Descent Type on GCR ranking (with random intercepts for Language Family) revealed that the kinship terminologies of unilineal societies rank GCR higher than bilateral societies ($\beta = 0.9068$ log-odds, $SE = 0.2478$, $p < 0.001$), corresponding to a 2.47 times increase in the odds of a higher ranking. A similar regression predicting Descent Type by Lineality weight indicated that bilateral societies rank Lineality higher than unilineal societies ($\beta = 0.5447$ log-odds, $SE = 0.2681$, $p = 0.0422$), corresponding to a 1.72 times increase in the odds of a higher ranking. This indicates that the ranking of a particular feature in a particular language is influenced by the descent patterns of the community, and therefore that patterns of descent influence the semantic distinctions a society will make in its kin terms – while nevertheless being shaped by the cross-linguistic cognitive biases in feature ranking that we have also identified here.

Second, we tested the claim that marriageable kin should be distinguished from unmarriedable kin (Murdoch, 1949; Coult, 1965; Goody, 1970). For example, there is a greater need to distinguish between your siblings and your cousins in societies where it is permitted to marry your cousins – a distinction by Lineality. If this were the case, we would expect that societies that permit marriage to any cousin will rank Lineality higher than societies that do not permit cousin marriage, reflecting that non-marriageable siblings need to be distinguished (by using distinct kin terms) from marriageable cousins. Some societies permit marriage only

to cross-cousins (children of a mother’s brother or father’s sister only) – making Gender of Connecting Relative the imperative distinction between marriageable and unmarried kin. In these societies, we would expect that GCR ranks higher than in societies who permit marriage to all cousins or no cousins, reflecting the importance of distinguishing the marriageable cross-cousins from the non-marriageable cousins. The idea is that if all cousins are marriageable or no cousins are marriageable, the need to distinguish by GCR is lessened.

These predictions are not supported in our data: an ordinal regression model predicting Lineality ranking by Cousin Marriage Allowance reveals no difference in ranking between societies who permit marriage to cousins and those who do not ($\beta = 0.05438$ log-odds, $SE = 0.21565$, $p = 0.801$). A second model predicting GCR ranking by Cross-Cousin Marriage Allowance also reveals no difference in ranking for societies who permit cross-cousin marriage only vs. no cousin marriage at all ($\beta = -0.104$, $p = 0.9196$), nor between cross-cousin marriage and other kinds of cousin marriage ($\beta = -0.965$, $p = 0.1161$). Taken together, these results indicate that the kin marriage practices of a society are not reliably reflected in the semantics of its kinship terminology — the relative ranking of features is not affected by whether or not cousin marriage is permitted, or by which kinds of cousins are marriageable. Contrary to longstanding claims in kinship anthropology, our results here suggest that while kin semantics *may* reflect social practice, not all aspects of social organisation reliably predict kin term semantics.

3. Discussion

We have shown that individual languages differ from one another in the semantics they encode with their kinship terminology. We also found that the features themselves vary in the extent to which they are encoded across languages, constraining the space of possible kinship semantics. Some features are encoded robustly across kin types and are highly frequent (like Generation), while other features are encoded in only a few kin types and are much rarer (like Speaker Gender).

We also found that there is a close correspondence between this semantic feature hierarchy and the behaviour of participants extending kinship terms to new referents in a controlled experiment: participants preferred to extend terms to kin who share features that are ranked higher cross-linguistically, meaning that higher-ranked features were more likely cues to shared category membership. We suggest that this bias towards higher-ranked features drives the cross-linguistic distribution of kin semantics; accordingly, features like Generation and Gender are encoded the most frequently in natural languages.

Our participants were all native English speakers, so their behaviour could be driven by their native language, rather than a universal cognitive bias. However, there are several reasons to believe this is not the case. Our experimental task aimed to reduce native-language transfer as much as possible: no English kin terms were used in the critical trials, and though the superordinate terms ‘parent’, ‘sibling’, and ‘child’ were used to familiarise participants with how to read the tree diagram, there was no critical trial where it was possible for participants to extend a term such that it corresponded to exactly the meanings of those

English terms. Furthermore, the critical trials often could not be solved with an English-like categorisation, as participants were forced to classify kin that do not share a term in English. Finally, and most importantly, English-speaking participants are actually a good test of the phenomenon we are interested in, as English does not follow the hierarchy perfectly (see Figure 3), since it ranks Lineality above Gender and does not encode GCR at all. The fact that our English speakers replicate the hierarchy would therefore be quite surprising if they were simply transferring kin term semantics from their native language. Although it is possible that our participants may have rarely classified by GCR simply because English does not encode GCR, experimental research with English-speaking children shows that they do understand the semantic distinction between ‘mother’s side’ and ‘father’s side’, even though it is not made explicit terminologically (Steele & Thomas, 2025).

On the basis of these results, we therefore claim that people may have a cognitive bias that privileges certain features like Generation and Gender over other features like Lineality and GCR, and (more tentatively) that this bias goes beyond language-specific experience. We propose that over time, this bias shapes the cross-linguistic distribution of kin term semantics.

More specifically, we argue that our experiment results can be explained by a cognitive bias for more readily identifiable semantic features. Generation and Gender only require knowledge of the referent themselves; they are *intrinsic* to a particular individual (Langton & Lewis, 2014). On the other hand, features like Lineality, GCR, or Seniority are not physically identifiable, and can only be derived through shared relationships; they are *relational*, requiring additional knowledge about how you and the referent are related to some third individual (Gentner & Kurtz, 2005). For example, to identify the GCR value of an relative requires that you also know the Gender value of the relative who connects you to them. In general, categories formed by shared relational properties are known to be harder to identify (Gentner & Kurtz, 2005; Asmuth & Gentner, 2017) and also harder to learn (Gentner, 2005; Corral et al., 2018) than intrinsic properties. For instance, children learning kin terms commonly misinterpret relational features as intrinsic ones, but not vice-versa; they preferentially define a grandfather as a kindly old man, not specifically the father of a parent (Keil & Batterman, 1984). They may also overextend intrinsic features, making errors like overextending the term *brother* to boys in general (Blythe et al., 2020). This kind of overgeneralisation may lead to intrinsic features being used more prolifically within the kinship system, explaining their higher position on the hierarchy.

Our experiment results also suggest that relational features are less readily identifiable. While they generally preferred to classify kin by higher-ranked features, our participants performed around chance in the Lineality-GCR trial. This suggests either that they do not have a preference for Lineality over GCR as basis for classification, or that they struggled to identify the relevant feature distinctions in these trials. We therefore argue that the cognitive bias displayed by our participants is in fact a bias for simplicity – given the evidence they have received, participants favour simpler hypotheses about the meaning of words (cf. Markham et al., 2024; F. Xu & Tenenbaum, 2007). Intrinsic features that are simpler and easy to identify top the hierarchy, and we suggest that the more complex relational features are encoded less often *in virtue* of the fact that they

are hard to identify.

The pressure to encode salient intrinsic features may be modulated by a pressure for kinship terminologies to be informative in a particular social niche, favouring encoding distinctions that are communicatively relevant, and underspecifying on communicatively less-important dimensions (e.g. S. Kirby et al., 2015; Carr et al., 2020; Silvey et al., 2015; Winters et al., 2018; Kemp & Regier, 2012; Kemp et al., 2018; Rosch, 1978). In addition to a cognitive bias for certain features driving the ranking of semantic features cross-linguistically, our cross-cultural analyses show that variation between languages in the ranking of features *may* be conditioned by social practice. While Lineality may be generally more commonly used as a means to distinguish kin, GCR may win out if there is a communicative need to distinguish the gender of a connecting relative – i.e. if a society is structured by matriliney or patriliney. Conversely, Lineality will be upweighted if there is a need to distinguish kin in the direct line – i.e. if a society is bilateral. Social organisation, like who you descend from, requires an understanding of relationships between individuals. To the extent that we find social practice corresponds with kinship semantics, this means that encoding relational properties in language *may* provide useful cues about the social niche.

However, although we confirm long-standing predictions about how kin terms reflect social norms around descent (Murdock, 1949; Coult, 1965), we find no evidence for the claim that kin term semantics reflect cousin marriage practices (Goody, 1970; Murdock, 1949). This resonates with other work which suggests that some commonly-claimed relationships between kin terms and social practice do not hold up to statistical scrutiny (Passmore & Jordan, 2020; Guillon & Mace, 2016). However, it is perhaps surprising, given that the relationships between kin terms and cousin marriage practices are some of the few correlations that have been shown to survive even when accounting for genetic and geographical effects (Passmore & Jordan, 2020). We note that research on the relationship between kinship terms and social practice has historically relied on a ‘typologised’ view of kin term variation, one that divides kinship systems into one of six types (Murdock, 1949) – but here, we have access to data at the kin term level. Indeed, more recent work using the Kinbank data to more accurately characterise kinship variation finds no support for an association between kin terms and cousin marriage (for Bantu languages) (Passmore et al., 2023). Similarly, our analyses use data which identify the critical semantic distinctions, and we do not find support for the association between kin terms and cousin marriage. We therefore propose that the more we characterise languages using data at the kin term level (rather than at a holistic, typological level), the more fragile these purported relationships become.

Overall, these results suggest that cultural communicative need is not a reliable predictor of the kinds of distinctions we should expect to see in a kinship system. Instead, we emphasise our experimental results, which indicate that the most reliable predictor of kin semantics is cognition. That is, Generation and Gender top the hierarchy by a considerable margin – and we argue that this is in response to a cognitive bias for simpler, salient intrinsic features. While individual languages may vary their semantics in response to social practice, this effect is considerably more variable.

By employing assumptions about topographicity in form-meaning mappings, we found that languages

vary in the features they encode, and the features themselves vary in the extent to which they tend to be encoded. We proposed a hierarchy of semantic encoding for kinship: Generation > {Gender > Lineality > Gender of Connecting Relative} > {Seniority > Speaker Gender}. On the basis of experimental evidence and cross-cultural comparison, we suggest that the hierarchy reflects a trade-off between a cognitive bias for kin terms to encode simple intrinsic features, but also to be informative of the social organisation of kin in the community. These competing pressures have been shown to constrain the space of possible variation in kin categorisation (Kemp & Regier, 2012), as well as numerous other semantic domains (Kemp et al., 2018; Zaslavsky et al., 2018; Y. Xu et al., 2016; Khetarpal et al., 2009). Here, we propose that these two pressures also compete to shape kin terms at the semantic level. Our findings contribute to the growing canon on how kinship terminology is structured for efficient communication.

4. Methods

The following sections give a detailed methodological description of the three studies reported here: the typological model, the experiment, and the cross-cultural analysis. Materials for all studies can be viewed at [LINK REDACTED FOR PEER REVIEW], including all data, all model and analysis code, the experiment stimuli, and experiment scripts. The design, sample size, exclusion criteria and analysis plan for the experiment were preregistered. For the curious reader, a miniature version of the experiment task can be completed at [LINK REDACTED FOR PEER REVIEW].

4.1. Model

We exploit the expected topographic relationship between form and meaning – i.e., that more similar wordforms have more similar meanings – to model how six semantic features are distributed across kin term semantics in 1242 languages. These six semantic features are Generation, Gender, Lineality, Gender of Connecting Relative, Seniority, and Speaker Gender. We measure topographicity as the correlation between semantic similarity and form similarity. In order to determine which semantic features are encoded by a particular language, we ask: in this language, which feature weightings would lead to the maximal correlation between forms and meanings? Here, we describe our measures of semantic similarity and form similarity, and how we manipulated semantic similarity to produce the maximal correlation between these two measures.

Data. Our data on kinship terminology come from Kinbank, a database of kinship terminology spanning 1229 languages across 21 major language families (e.g. discriminated at the level of Indo-European, Austronesian, or Atlantic-Congo; Passmore et al., 2023). In this database, each kin term is associated with an objective, language-independent meaning, which we refer to as a **kin type**. These meanings incorporate a number of defining features for each kin type, such as their gender, their connecting relative, or their age relative to another individual. For example, in English, the kin term *uncle* is associated with (at least) four distinct kin types: ‘mother’s elder brother’, ‘mother’s younger brother’, ‘father’s elder brother’, and ‘father’s younger brother’. While we do not distinguish these kin types in English, they are nonetheless distinct in

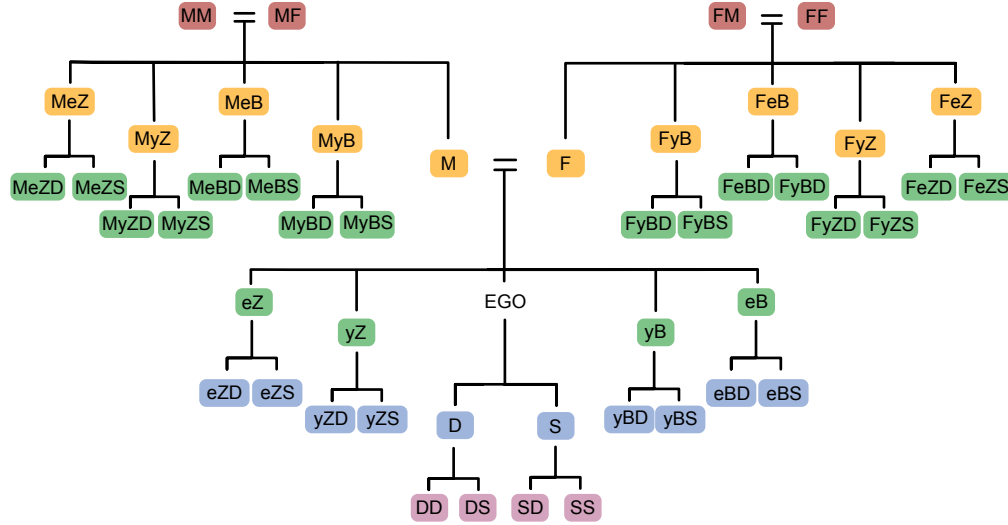


Figure 7: A schematic of the kin types we included in our analysis. ‘Ego’, at the centre, represents the reference individual. Shorter branch length indicates elder individuals, longer branch length indicates younger. Relative age is variably encoded, so only some kin types are marked for the elder/younger distinction. Kin types are given in shorthand: e.g. ‘M’ = mother, ‘FyB’ = father’s younger brother, ‘eZ’ = elder sister, ‘DS’ = daughter’s son, etc. Colours indicate kin types in the same generation for clarity. All kin types pictured here were included from the male and female speaker perspective.

other languages – such as Myene (Atlantic-Congo), which uses the terms *ngw’onèró*, *ngw’erumbe*, *rer’onèró* and *rer’erumbe* respectively. By comparing whether these language-independent meanings have the same or similar terms, we can get a sense for which semantics are relevant in a particular language’s kinship terminology: in English, the term *uncle* collapses distinctions of GCR and Seniority, while those distinctions are made explicit in Myene.

For our analysis, we considered 48 kin types spanning five generations: grandparents, parents and their siblings, siblings and cousins, children and niblings, and grandchildren. To keep the meaning space within a reasonable scope, we only looked at consanguineal kin (blood-related family), not affinal kin (those related by marriage) or fictive kin (social ties that are neither genealogical nor marital). For instance, our model does not consider ‘parent’s sibling’s husband’ or ‘parents’ male friend’ as possible meanings of *uncle* in English. Figure 7 gives a full schematic of the kin types we included in our analysis.

Measuring semantic similarity. To determine the semantic content of a kin type, we created a feature matrix for each kin type in Figure 7 by coding their value for each of the six semantic features. The full coding scheme for each feature is given in Table 2. For example, ‘man’s mother’s elder sister’ is coded G^{+1} for Generation, *woman* for Gender, *collateral* for Lineality, *woman* for GCR, *elder* for Seniority, and *man* for Speaker Gender. Not all kin types have a value for all features; when this was the case, we assigned a kin type an NA value for that feature.

We calculated the semantic similarity between a pair of kin types (x, y) by comparing their feature matrices. For each feature f_i in the set of features F we assigned a similarity score for individuals x and y based

Feature	Description	Values	Notes
Generation	Which generation the kin type belongs to.	G^{+2} , G^{+1} , G^0 , G^{-1} , G^{-2}	G^{+2} = Ego's grandparents G^{+1} = Ego's parents G^0 = Ego's generation G^{-1} = Ego's children G^{-2} = Ego's grandchildren
Gender	Gender of kin type.	woman, man	
Lineality	Whether kin type is in Ego's direct lineage.	lineal, collateral	'Direct lineage' includes grandparents, parents, children or grandchildren.
Gender of Connecting Relative	Gender of the individual who links Ego to kin type.	woman, man, NA	e.g. mother's sister is linked by mother NA values are assigned to Ego's direct descendants, as they are not linked by another individual.
Seniority	Whether kin type is elder or younger relative to their counterpart.	elder, younger, NA	e.g. elder brother vs younger brother NA values are assigned when there is no counterpart with whom to compare age (e.g. grandparents).
Speaker Gender	Gender of Ego, the person using the term.	woman, man	

Table 2: Full coding scheme for semantic features of kin. Each kin type is assigned a value for each of the six features. When a feature does not apply to a kin type, they are assigned NA for that feature (e.g. Seniority is undefined for parents).

on whether or not they shared a value for that feature. If the pair differed on the feature, $f_i(x, y) = -1$; if the pair shared the feature, then $f_i(x, y) = 1$. If one or both kin types in the pair had an NA value for the feature, then $f_i(x, y) = 0$. The semantic similarity between x and y is then a weighted sum of their by-feature similarities:

$$Sem(x, y) \equiv \sum_{i \in F} f_i(x, y) w_i \quad (1)$$

where w_i is the weight for feature f_i , with weights lying between 0 and 1 and normalised such that $\sum_{i \in F} w_i = 1$.

This weight determines the relative importance of the semantic feature f_i in determining similarity between individuals in the kinship system. If all semantic features were equally important, then $w_n = 1/(|F|)$. However, we already know that not all languages encode all features equally – in English, we make dis-

inctions by Generation (*mother* vs *grandmother*), Gender (*aunt* vs *uncle*) and Lineality (*brother* vs. *cousin*), but not GCR, Seniority, or Speaker Gender. We want the values of the weights to reflect this disparity. For example, if a language were to place a high weight on Gender, that would mean that individuals of the same gender would share a term more often than individuals differing in gender. Or, if Gender were to have a weight of 0, that would indicate that similarity or difference in gender did not contribute at all to whether two individuals share a kin term.

The values of these weights are unknown; our goal is to search for their optimal values such that topographicity, the correlation between pairwise semantic similarity and form similarity, is maximised.

Measuring form similarity. To measure how similar a pair of kin terms are in their form, we measured their normalised Levenshtein edit distance, E : the number of single-letter insertions, deletions, or replacements needed to turn one string into another, proportional to the length of the longer string (Levenshtein, 1965). To make this a similarity measure, we took form similarity to be $1 - E$.

For each pair of kin types, we measured the form similarity between the terms used to refer to those types. For example, the kin types ‘mother’s mother’ and ‘father’s mother’ are both referred to with the term *grandmother* in English; these strings require no edits as they are identical ($E = 0$; $1 - E = 1$). On the other hand, the kin types ‘mother’s elder sister’ and ‘mother’s elder brother’ are referred to with the terms *aunt* and *uncle* respectively; to turn one string into the other requires four edits, insertions or deletions ($E = 0.8$; $1 - E = 0.2$). Since kin terms are different in different languages, the form similarity measure for each pair differs between each language.

Measuring topographicity. We measured the topographicity of a kinship system by calculating the Pearson correlation between pairwise semantic similarity and pairwise form similarity (cf. S. Kirby et al., 2008; Monaghan et al., 2014). For each pair of distinct kin types in Figure 7, we computed their semantic similarity and form similarity, and calculated the correlation between the resulting distribution of pairwise semantic similarities and pairwise form similarities. This correlation tells us the extent to which similar meanings map to similar forms, and dissimilar meanings map to dissimilar forms. We searched for the set of weights w_i across all semantic features that would maximise this correlation in a particular language. Essentially, we assume that a language topographically maps the semantic space of the speakers of that language, and use this procedure to discover the structure of that semantic space.

Finding optimal feature weights. We wanted to know whether different languages encode different semantic features, and if so, how much languages differ from each other. To do this, we manipulated the measure of semantic similarity, searching for the set of weights that would produce the best correlation with form similarity for each language in our dataset. We used a gradient ascent algorithm to optimise our semantic

similarity measure for a particular language¹. Gradient ascent searches for the maximum output of an objective function by calculating the direction of steepest ascent towards the maximum, then searching for an appropriate step size to make along that gradient given the data, and iteratively making incremental changes to the parameter weights until it converges on the maximum possible value of the function.

In our case, the objective function that we optimised is the topographicity metric, the correlation between all pairwise form similarities and all pairwise semantic similarities. The parameters are the weights on each semantic feature. The algorithm is initialized with a random set of weights, then calculates the direction of the greatest possible correlation between form and meaning. Then, it iteratively updates the weights associated with each semantic feature until the semantic similarity metric cannot be changed in such a way as to increase the correlation with the form similarity metric.

This leaves us with a set of optimal feature weights for each language — optimal in the sense that gradient ascent has been unable to discover any which are more informative about which how similar kin terms will be to one another. We take the weights to represent the extent to which each feature is encoded in the language’s kinship terminology. A higher weight indicates that sharing this feature is highly correlated with sharing parts of form, and a lower weight indicates that sharing that feature has little or no impact on sharing form.

4.2. Experiment

Our experiment was approved by the [DEPARTMENT REDACTED] ethics committee at [INSTITUTION REDACTED] (ref. no. 284-2425/1). All participants gave informed consent to participate and to have their data anonymised and analysed.

4.2.1. Procedure and Materials

The experiment was implemented with jsPsych (de Leeuw et al., 2023), and was conducted remotely through an interface that ran online in participants’ web browsers.

In the critical trials of the experiment, participants had to assign two labels to three kin types, indicating their preference for who should be classified together. Each participant completed six critical trials, one that tested their preference for classifying along each pair of four semantic features: Generation-Gender, Generation-Lineality, Generation-GCR, Gender-Lineality, Gender-GCR, and Lineality-GCR. We chose to test these four features as they are easy to represent visually, either by position on a family tree diagram (Generation, Lineality, GCR) or by the visual characteristics of an individual (Gender). Seniority and Speaker Gender were excluded because they are more difficult to represent on-screen, and we therefore felt that participants

¹Gradient ascent was implemented in Python using the `optimise.minimize` method from the `scipy` package (Virtanen et al., 2020), with `method = L-BFGS-B` and `bounds = (0,1)`. We used the L-BFGS-B algorithm for two key reasons. First, it permits bounds on the possible values of the parameters, and we wanted these to remain between 0 and 1 and to sum to 1, to facilitate comparison between features and languages in our analysis. Second, L-BFGS-B is appropriate for objective functions that are differentiable and smooth (which our objective function, Pearson correlation, is).

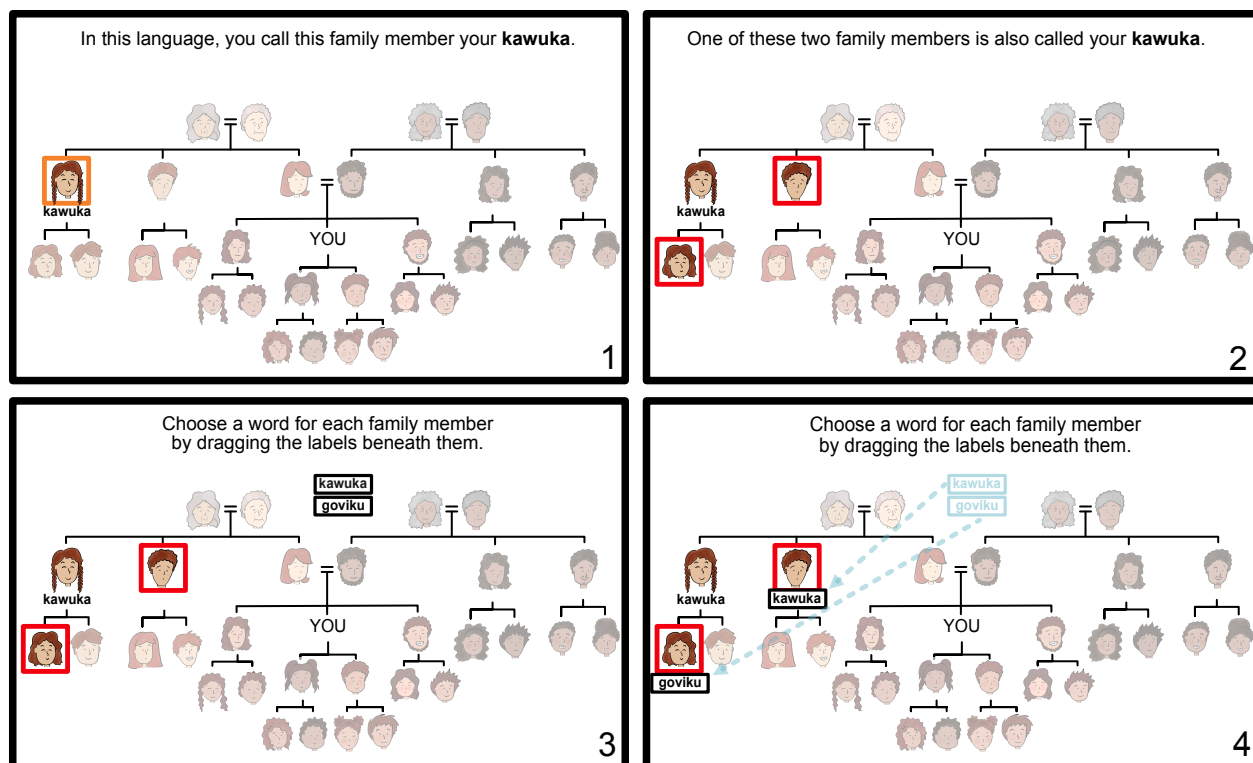


Figure 8: Depiction of the experimental task for a Generation-Gender trial. In the first panel, the participant is shown a kin term in a novel language referring to a focal individual in the tree – e.g. *kawuka* for ‘mother’s sister’. In the second panel, two more individuals are highlighted and participants were informed that one of them is also referred to as *kawuka*. Crucially, they differ from the focal individual along two feature dimensions: ‘mother’s brother’ differs by Gender but shares Generation and ‘mother’s sister’s daughter’ differs by Generation but shares Gender. In the third panel, the participant must choose to extend the kin term *kawuka* to one of the two individuals, and give the other a novel term, *goviku*. In the fourth panel, the participant has dragged *kawuka* to ‘mother’s brother’ and *goviku* to ‘mother’s sister’s daughter’, indicating a preference to classify kin according to shared Generation, not shared Gender.

might struggle to identify them as the relevant semantic dimension. The order of trials was randomised per participant.

We exemplified kin types using a family tree diagram populated by illustrated characters. Before the critical trials, participants were given a brief demonstration of how to read this family tree diagram. Participants were told to imagine themselves at the centre of the family tree diagram, and they could click to reveal other individuals in the family tree one group at a time. Their relationships to these individuals were described using the English terms *parent*, *sibling* and *child*: e.g. “These are the parents of your parents”. The full tree remained on screen during all trials so that participants did not have to remember information about which characters represented which relationships. The illustrated characters populating the family tree diagram were designed such that closely related individuals shared visual characteristics like hair colour, hair style, and skin tone. The orientation of the tree was randomised between participants; each participant was randomly assigned to one of four conditions determining whether the mother or father were on the right- or left-hand side of the tree, and determining the left-to-right order of men and women.

In each critical trial, participants were presented with the family tree diagram with all characters greyed out. Then, one individual was highlighted with a yellow box (the exemplar) and a kin term would appear beneath them, e.g. *kawuka*. A prompt also appeared: e.g. “In this language, you call this family member your **kawuka**”. Then, two more individuals (the targets) were highlighted with red outlines, and the prompt would read, “One of these two family members is also called your **kawuka**.” Finally, two draggable labels would appear at the top of the screen, one reading *kawuka* and the other an alternative kin term, e.g. *goviku*. Participants were then instructed to choose a word for each target by dragging and dropping the labels beneath them. The label would snap into place beneath each target, meaning participants could not label any other non-target kin. They could move the labels around as much as they wished, but they could only continue to the next trial once both labels had been assigned. In each of the six trials, we designed the artificial kin terms with differing syllable structure and diacritics, to emphasise that the kin terms in each trial were from a different language and discourage participants from using information from previous trials to inform their response. Which terms appeared in which trials was randomised per participant.

Crucially, the two targets differed from the exemplar along the critical pair of feature dimensions, but had identical values for the other features, forming a ‘minimal triad’. For example, Figure 8 depicts a Generation-Gender trial where the exemplar is ‘mother’s sister’ and the two targets are ‘mother’s brother’ (differs by Gender, but not by Generation) and ‘mother’s sister’s daughter’ (differs by Generation, but not by Gender). By choosing to give the label *kawuka* to ‘mother’s brother’ and *goviku* to ‘mother’s sister’s daughter’, the participant would be revealing a preference to classify together kin who share their generation over classifying kin who share their gender. For each trial type, there were numerous triads of kin types that could exemplify the relevant feature distinctions. For example, a minimal triad between Generation-Gender can be instantiated by ‘mother’s sister’, ‘mother’s brother’, and ‘mother’s sister’s daughter’ (as in Figure 8) but also by ‘father’s brother’, ‘father’s sister’, and ‘father’s brother’s son’, among many other combinations. In each trial, a minimal triad was selected at random from the full list of possible triads, to exclude the possibility that participants’ classification behaviour had to do with specific kin types rather than semantic features.

Each participant completed a single attention check trial, which also doubled as a check to ensure they could read the family tree properly. This trial was completed after the six critical trials and was the same for all participants. For this trial only, we used English kin terms: ‘brother’s son’ was the exemplar, labelled with the term *nephew*. The targets were ‘mother’s brother’s son’ and ‘sister’s son’, and the two draggable labels were *nephew* and *cousin*. If a participant responded incorrectly to this trial, we excluded all their data from analysis as this indicated that they were either not paying attention, or could not correctly read the family tree diagram.

4.2.2. Participants

We recruited 115 participants from Prolific who were self-reported native English speakers living in the UK or US and who had an approval rating of 98% or higher. The experiment lasted approximately 10 minutes, and participants were paid £2.05 for their participation (in line with UK National Minimum Wage).

Feature	Values	Original D-PLACE values
Descent type	Unilineal	Matrilineal, Patrilineal
	Bilateral	Bilateral, Ambilineal, Mixed
Line of descent	Matrilineal	Matrilineal
	Patrilineal	Patrilineal
Cousin marriage permitted	All cousins	Any first cousins
	Some cousins	Matrilateral cross only, Patrilineal cross only, Cross-cousin, Only second cousins, Some second only, Maternal only, Paternal only
		No first/second cousins, No first cousins
Cross-cousin marriage permitted	Cross-cousins only	Cross-cousin, Matrilateral cross only, Patrilineal cross only
	Other cousins only	Only second cousins, Some second cousins, Maternal only, Paternal only, Any first cousins
	No cousins	No first/second cousins, No first cousins

Table 3: Coding scheme for data on descent type and marriage practices. ‘Values’ indicates our coding scheme; ‘Original D-PLACE values’ indicates which values from the existing data in D-PLACE informed each code.

15 participants were excluded for failing an attention check, leaving us with 100 participants’ data usable for analysis.

4.3. Cross-cultural analysis

Our data on the rank for each semantic feature are derived from the output of our computational feature-weight optimisation. For each language, we calculated the relative rank of each feature, according to the weights assigned by the optimisation. Higher ranks indicate that more weight is placed on that feature. For example, if a language’s highest weight was on Generation, then Generation would receive a rank of 6. If a language’s lowest weight was Speaker Gender, then Speaker Gender would receive a rank of 1. If two features were weighted 0 – i.e. if neither contribute to the mapping between forms and meanings for that language – then both features were ranked 1, and the next highest weighed feature received rank 3. This was done so that languages that encode fewer features were not penalised, so their highest ranked feature still received rank 6.

Our cross-cultural data on social practice comes from D-PLACE (K. R. Kirby et al., 2016). We used the Major Descent Type variable (EA043, originally from the Ethnographic Atlas: Murdock et al., 1999) to code whether societies are unilineal or bilateral and whether they are matri- or patrilineal, and the Cousin Marriage

Permitted variable (EA023, originally from the Ethnographic Atlas: Murdock et al., 1999) to code a society's acceptance of cousin marriage. We re-coded the data according to coding scheme given in Table 3.

After combining the kin term semantic data with the D-PLACE data, we were left with 432 languages and societies for which we had descent type data (215 bilateral and 217 unilineal), 364 languages and societies for which we had cousin marriage data (24 all cousins permitted, 128 some cousins permitted, 212 no cousins permitted) and 359 for which we had cross-cousin marriage data (90 cross cousins permitted, 57 other cousins permitted, 212 no cousins permitted). We used cumulative link mixed effects models to predict the effect of social practice on whether a feature receives a higher rank.

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